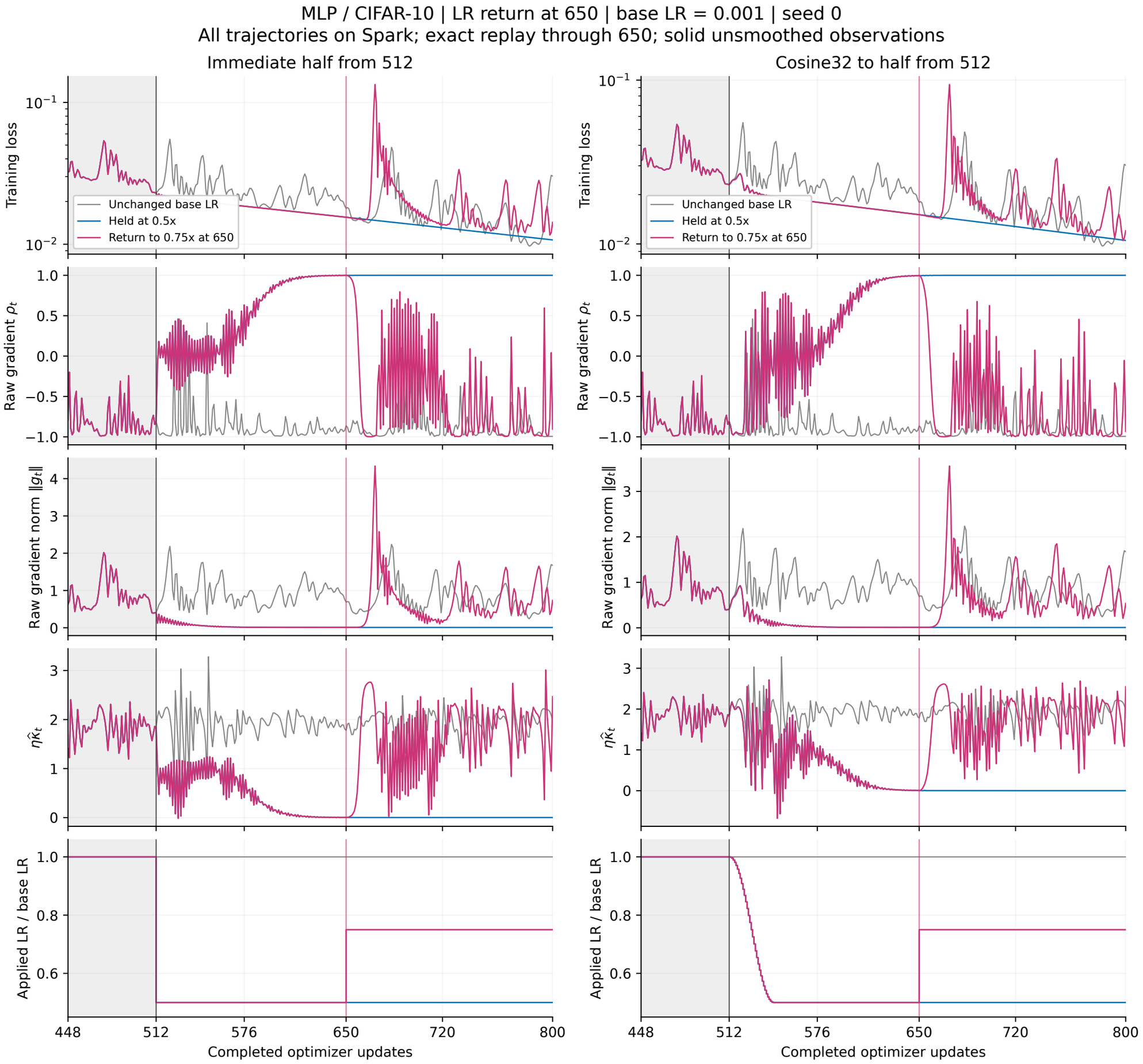

MLP / CIFAR-10 | LR return at 650 | base LR = 0.001 | seed 0
All trajectories on Spark; exact replay through 650; solid unsmoothed observations
Immediate half from 512
Cosine32 to half from 512
Training loss
Unchanged base LR
Held at 0.5x
Return to 0.75x at 650
Raw gradient ρ_t
Raw gradient norm $\|g_t\|$
$\eta\hat{\kappa}_t$
Applied LR / base LR
Completed optimizer updates